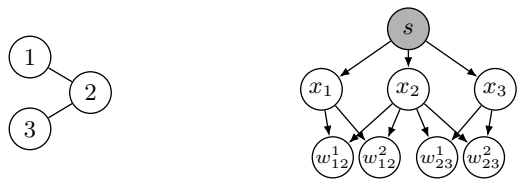
APPENDIX

A. Proof of Theorem 3.2

We reduce from DkS: given an undirected graph $H = (V_H, E_H)$ and an integer k, find a k-vertex set $T \subseteq V_H$ that maximizes $e_H(T)$, the number of edges with both endpoints in T. Let $d^* = \max_{|T|=k} e_H(T)$ be the optimum.

Construction. From a DkS instance (H, k) we build, in polynomial time, an IMEB instance: a graph $G = (V, E)$ with a single seed $S = \{s\}$ and the same budget k. Figure 5 illustrates the construction, which arranges G into three layers. At the top sits the seed s. In the middle, each vertex $i \in V_H$ becomes a *selector* x_i, joined to the seed by an edge $g_i = \langle s, x_i \rangle$. At the bottom, each edge $\{i, j\} \in E_H$ becomes M *witnesses* $w_{ij}^1, \ldots, w_{ij}^M$ for an integer M fixed later. Each witness receives exactly two incoming edges, one from x_i and one from x_j, the selectors of the endpoints of $\{i, j\}$. Moreover, all edge probabilities are set to 1, thus $\rho(B)$ equals exactly the number of nodes that blocking B disconnects from s.



(a) DkS instance H (b) Constructed graph G

Fig. 5. The reduction from DkS to IMEB with $M = 2$.

Optimal value. We first determine $\rho(B^o)$, the optimal IMEB objective on the constructed instance. Let $B_T = \{g_i : i \in T\}$ be the blocker corresponding to a k-subset $T \subseteq V_H$. Blocking B_T has two effects. First, it disconnects the k selectors $\{x_i : i \in T\}$ from s. Second, since each witness is reachable from s only through its two selectors x_i, x_j, B_T also removes all M witnesses of every edge with both endpoints in T. Combining the two, $\rho(B_T) = k + M\, e_H(T)$, which equals $k + M\, d^*$ for an optimal T. No blocker exceeds this value: take any blocker B with $|B| \le k$ and set $T_B = \{i : g_i \in B\}$. Its selector in-edges $\{g_i : i \in T_B\}$ disconnect $|T_B| + M\, e_H(T_B)$ nodes, while each of the remaining $|B| - |T_B|$ blocked edges, a witness in-edge, disconnects at most one further witness. Hence $\rho(B) \le |B| + M\, e_H(T_B) \le k + M\, d^*$ and it holds that $\rho(B^o) = k + M\, d^*$.

Gap preservation. Without loss of generality $d^* \ge 1$, since otherwise no k-subset of H spans an edge and the DkS optimum is found in polynomial time. Let $\mathcal{A}$ be any α-approximation for IMEB, fix $\eta > 0$, and set $M = \lceil k/\eta \rceil$, so that $k/M \le \eta$. On G, $\mathcal{A}$ returns a blocker B with $\rho(B) \ge \alpha\, \rho(B^o) = \alpha(k + M\, d^*)$. From B we read off the DkS solution T_B. Combining the bound $\rho(B) \le k + M\, e_H(T_B)$ with $\rho(B) \ge \alpha(k + M\, d^*)$, we have $M\, e_H(T_B) \ge \alpha\, M\, d^* - (1 - \alpha)k$, hence

$$e_H(T_B) \;\ge\; \alpha\, d^* - \tfrac{(1-\alpha)k}{M} \;\ge\; \alpha\, d^* - \eta \;\ge\; (\alpha - \eta)\, d^*,$$

where the last two steps use $k/M \le \eta$ and $d^* \ge 1$. Thus an α-approximation for IMEB gives, for every $\eta > 0$, an $(\alpha - \eta)$-approximation for DkS: with $\alpha = 1 - \varepsilon$ it turns a PTAS for IMEB into a PTAS for DkS. The theorem follows since DkS admits no PTAS unless $\mathrm{NP} \subseteq \bigcap_{\epsilon > 0} \mathrm{BPTIME}(2^{n^\epsilon})$ [37]. □

B. Proof of Lemma 3.3

Since blocking additional edges can only reduce the set of reachable nodes from S in any realization, the monotonicity of $\rho(\cdot)$ follows. We prove the non-submodularity and non-supermodularity of $\rho(\cdot)$ via a counterexample illustrated in Fig. 1(a), where the seed set $S = \{v_0\}$ and all edge probabilities are set to 1. Let $A_1 = \emptyset$ and $A_2 = \{\langle v_0, v_2 \rangle\}$, and consider adding $\langle v_0, v_1 \rangle$. Blocking $\langle v_0, v_1 \rangle$ alone does not disconnect any node, since v_1 remains reachable via $v_0 \to v_2 \to v_1$, so $\rho(\{\langle v_0, v_1 \rangle\}) = 0$. However, when $\langle v_0, v_2 \rangle$ is already blocked, v_2 is disconnected and the alternative path to v_1 is destroyed; blocking $\langle v_0, v_1 \rangle$ additionally disconnects v_1, giving $\rho(\{\langle v_0, v_1 \rangle, \langle v_0, v_2 \rangle\}) = 2$. Hence

$$\begin{aligned}\rho(A_2 \cup \{\langle v_0, v_1 \rangle\}) - \rho(A_2) &= 1 \\ &> 0 = \rho(A_1 \cup \{\langle v_0, v_1 \rangle\}) - \rho(A_1),\end{aligned}$$

violating the submodularity. In addition, let $A_3 = \{\langle v_1, v_4 \rangle, \langle v_2, v_4 \rangle\}$ and $A_4 = A_3 \cup \{\langle v_0, v_3 \rangle\} \supseteq A_3$, and consider adding $\langle v_3, v_4 \rangle$. With A_3 blocked, v_4 is still reachable via $v_0 \to v_3 \to v_4$; adding $\langle v_3, v_4 \rangle$ eliminates this last path, disconnecting v_4, so $\rho(A_3 \cup \{\langle v_3, v_4 \rangle\}) - \rho(A_3) = 1 - 0 = 1$. With A_4 blocked, $\langle v_0, v_3 \rangle$ is removed, making v_3 unreachable and destroying the path $v_0 \to v_3 \to v_4$; consequently v_4 is already unreachable under A_4, so adding $\langle v_3, v_4 \rangle$ has no further effect: $\rho(A_4 \cup \{\langle v_3, v_4 \rangle\}) - \rho(A_4) = 2 - 2 = 0$. Hence

$$\begin{aligned}\rho(A_3 \cup \{\langle v_3, v_4 \rangle\}) - \rho(A_3) &= 1 \\ &> 0 = \rho(A_4 \cup \{\langle v_3, v_4 \rangle\}) - \rho(A_4),\end{aligned}$$

violating the supermodularity. □

C. Proof of Lemma 3.4

The submodularity ratio [38] is a metric to assess how closely a function approximates being submodular. For $A_1, A_2 \subseteq E$, the submodularity ratio of $\rho(\cdot)$ is the largest scalar $\gamma \ge 0$ such that

$$\textstyle\sum_{e \in A_1 \setminus A_2} \rho(e \mid A_2) \ge \gamma\big(\rho(A_2 \cup A_1) - \rho(A_2)\big). \qquad (12)$$

Consider the graph in Fig. 1(a), where all edge probabilities are 1 and $S = \{v_0\}$. Let $A_1 = \{\langle v_0, v_1 \rangle, \langle v_2, v_1 \rangle\}$ and $A_2 = \emptyset$. Then the left side of Eq. (12) is $(\rho(\{\langle v_0, v_1 \rangle\}) - \rho(\emptyset)) + (\rho(\{\langle v_2, v_1 \rangle\}) - \rho(\emptyset)) = 0 + 0 = 0$, since blocking either edge alone does not disconnect v_1 (the alternative path $v_0 \to v_2 \to v_1$ or $v_0 \to v_1$ remains intact). The right side is $\gamma \cdot (\rho(\{\langle v_0, v_1 \rangle, \langle v_2, v_1 \rangle\}) - \rho(\emptyset)) = \gamma \cdot 1$, since blocking both edges simultaneously disconnects v_1. Thus $0 \ge \gamma$, which forces $\gamma = 0$. More generally, appending $n - 5$ additional nodes $v_5, \ldots, v_{n-1}$ with edges $\langle v_1, v_i \rangle$ for $5 \le i \le n - 1$ (all with probability 1) yields $0 \ge \gamma \cdot (n - 4)$, so $\gamma = 0$ regardless of n. This completes the proof. □

D. Proof of Lemma 4.2

Monotonicity is immediate since enlarging B can only introduce additional critical edges. For submodularity, define

$$f_\phi(B) = \sum_{v \in V \setminus S} \max_{e \in B} \mathbf{1}[e \text{ is critical for } v \text{ in } \phi].$$

Since $\rho^L(B) = \mathbb{E}_\phi[f_\phi(B)]$, it suffices to show that $f_\phi(\cdot)$ is submodular for each fixed realization ϕ, as a non-negative linear combination of submodular functions is submodular. For any $A_1 \subseteq A_2 \subseteq E$ and $e' \in E \setminus A_2$, the marginal gain of e' w.r.t. A_1 under a fixed ϕ is

$$\begin{aligned} &f_\phi(A_1 \cup \{e'\}) - f_\phi(A_1) \\ &= \big|\{v \in V \setminus S \mid e' \text{ is critical for } v,\ \nexists\, e \in A_1 \text{ critical for } v\}\big|, \end{aligned}$$

i.e., the number of nodes for which e' is critical and no edge already in A_1 is critical. Since $A_1 \subseteq A_2$, if some edge in A_1 is critical for v, then so is that same edge in A_2; hence the set of nodes contributing to the marginal gain can only shrink when A_1 is replaced by the larger set A_2, that is,

$$f_\phi(A_1 \cup \{e'\}) - f_\phi(A_1) \geq f_\phi(A_2 \cup \{e'\}) - f_\phi(A_2),$$

which completes the proof. □

E. Proof of Lemma 4.5

Monotonicity holds since enlarging B can only add more edges to $B \cap E(T_\phi)$, increasing the union. For the submodularity, since a nonnegative linear combination of submodular functions remains submodular, it suffices to show that $g_\phi(B) = \big|\bigcup_{\langle u,v\rangle \in B \cap E(T_\phi)} D_\phi(v)\big|$ is submodular for each ϕ. This is a coverage function: each edge $\langle u, v\rangle \in E(T_\phi)$ contributes the set $D_\phi(v)$, and $g_\phi(B)$ is the size of their union. For any $A_1 \subseteq A_2 \subseteq E$ and $e' = \langle u, v\rangle \in E \setminus A_2$, the marginal gain of e' is

$$g_\phi(A_1 \cup \{e'\}) - g_\phi(A_1) = \big|D_\phi(v) \setminus \bigcup_{\langle u',v'\rangle \in A_1 \cap E(T_\phi)} D_\phi(v')\big|.$$

i.e., the number of descendants of v not covered by A_1 in T_ϕ. Since $A_1 \subseteq A_2$, the set of covered nodes under A_2 contains that under A_1, so the marginal gain can only shrink:

$$g_\phi(A_1 \cup \{e'\}) - g_\phi(A_1) \geq g_\phi(A_2 \cup \{e'\}) - g_\phi(A_2).$$

This completes the proof. □

F. Proof of Lemma 5.1

We reduce from influence computation under the IC model, which is #P-hard [7]. Consider an instance of that problem: a graph G and a node $q \in V$ that has no in-neighbor, whose answer is $\sigma(\{q\})$. From this we construct G' by adding a new node r and an edge $\langle r, q\rangle$ of probability 1, and set $S = \{r\}$ and $B = \{\langle r, q\rangle\}$. For the lower bound, since $\langle r, q\rangle$ is the sole edge emanating from the seed r, in every realization ϕ it is live and its removal disconnects every node of $I_\phi(\{r\}) \setminus \{r\}$ from $S = \{r\}$. Hence $\langle r, q\rangle$ is critical for every such node, giving $\rho^L(B) = \mathbb{E}_\phi[|I_\phi(\{r\})| - 1] = \sigma(\{q\})$. For the upper bound, $\langle r, q\rangle$ is the only edge entering q in G', so $\langle r, q\rangle \in E(T_\phi)$ for every ϕ, with $D_\phi(q) = I_\phi(\{r\}) \setminus \{r\}$; thus $\rho^U(B) = \mathbb{E}_\phi[|I_\phi(\{r\})| - 1] = \sigma(\{q\})$. Hence, a polynomial-time algorithm for either bound would compute $\sigma(\{q\})$, contradicting the #P-hardness of influence computation. □

G. Proof of Lemma 5.3

Let $R_\phi(v)$ be the CE set of target v in realization ϕ.

$$\begin{aligned} \rho^L(B) &= \textstyle\sum_{v \in V \setminus S} \mathbb{E}_\phi[\max_{e \in B} \mathbf{1}[e \text{ is critical for } v]] \\ &= \mathbb{E}_\phi\Big[\textstyle\sum_{v \in V \setminus S} \mathbf{1}[B \cap R_\phi(v) \neq \emptyset]\Big] \\ &= \mathbb{E}_\phi\Big[\textstyle\sum_{v \in I_\phi(S) \setminus S} \mathbf{1}[B \cap R_\phi(v) \neq \emptyset]\Big] \\ &= \mathbb{E}[N(R) \cdot \mathbf{1}[B \cap R \neq \emptyset]]\,. \end{aligned}$$

The third equality drops the non-activated targets, for which $R_\phi(v) = \emptyset$. The last uses that the inner sum has $|I_\phi(S) \setminus S| = N(R)$ terms, hence equals $N(R)$ times the average of $\mathbf{1}[B \cap R_\phi(v) \neq \emptyset]$ over a target v drawn uniformly at random from $I_\phi(S) \setminus S$, which is the sampling of R in Definition 5.2. □

H. Proof of Lemma 5.7

Let $P_\phi(v)$ denote the AE set with target node v in realization ϕ. As Eq. (4) is structurally analogous to Eq. (2), the proof follows the same line of reasoning as Lemma 5.3:

$$\begin{aligned} \rho^U(B) &= \mathbb{E}_\phi\big[\textstyle\sum_{v \in V \setminus S} \mathbf{1}[B \cap P_\phi(v) \neq \emptyset]\big] \\ &= \mathbb{E}_\phi\big[\textstyle\sum_{v \in I_\phi(S) \setminus S} \mathbf{1}[B \cap P_\phi(v) \neq \emptyset]\big] \\ &= \mathbb{E}\big[N(P) \cdot \mathbf{1}[B \cap P \neq \emptyset]\big]. \end{aligned}$$

This completes the proof. □

I. Proof of Lemma 6.1

We use the martingale tail inequality: for a martingale $M_1, \ldots, M_x$ with $M_1 \leq a$ and $M_j - M_{j-1} \leq a$ ($j \in [2, x]$), and $\mathrm{Var}[M_1] + \sum_{j=2}^{x} \mathrm{Var}[M_j \mid M_1, \ldots, M_{j-1}] \leq b$, one has

$$\Pr[M_x - \mathbb{E}[M_x] \geq \lambda] \leq \exp\big(-\tfrac{\lambda^2}{\frac{2}{3}a\lambda + 2b}\big) \quad (\lambda > 0).$$

The increments X_i satisfy three properties: (i) $X_i \in [0, 1]$, since $0 \leq N(Q_i) \leq n^-$ and $\mathbf{1}[\cdot] \in \{0, 1\}$; (ii) $\mathbb{E}[X_i] = \mu$, by the unbiasedness of the estimator (Lemmas 5.3 and 5.7); and (iii) since $X_i \in [0, 1]$ gives $X_i^2 \leq X_i$, $\mathrm{Var}(X_i) = \mathbb{E}[X_i^2] - \mu^2 \leq \mathbb{E}[X_i] - \mu^2 = \mu(1 - \mu)$.

Now define $M_t = \sum_{i=1}^{t}(X_i - \mu)$ for $t = 1, \ldots, \theta$. The samples are drawn independently and each X_i depends only on Q_i, so $X_1, \ldots, X_\theta$ are mutually independent and $\mathbb{E}[X_t \mid X_1, \ldots, X_{t-1}] = \mu$; hence $\mathbb{E}[M_t \mid M_1, \ldots, M_{t-1}] = M_{t-1}$, so $(M_t)_{t \leq \theta}$ is a martingale with $\mathbb{E}[M_\theta] = 0$. By the same independence, the conditional-variance sum is $\sum_{k=1}^{\theta} \mathrm{Var}[M_k \mid M_1, \ldots, M_{k-1}] = \sum_{k=1}^{\theta} \mathrm{Var}(X_k) \leq \theta\mu(1 - \mu)$.

Applying the inequality to M_θ with $a = 1$ (since $M_j - M_{j-1} = X_j - \mu \leq 1$) and $b = \theta\mu$ gives

$$\Pr[M_\theta \geq \lambda] \leq \exp\big(-\frac{\lambda^2}{2\theta\mu + \frac{2}{3}\lambda}\big).$$

Since $\Lambda_{\mathcal{Q}}(B) = \theta\mu + M_\theta$, Eq (5) holds.

Next, apply it to the negated martingale $-M_\theta$. Since $X_i \geq 0$, both $-M_1 = \mu - X_1 \leq \mu$ and $-M_k - (-M_{k-1}) = \mu - X_k \leq \mu$, so $a = \mu$ and $b = \theta\mu(1 - \mu)$, giving

$$\Pr[M_\theta \leq -\lambda] \leq \exp\big(-\frac{\lambda^2}{2\theta\mu(1 - \mu) + \frac{2}{3}\mu\lambda}\big).$$

If $\lambda > \theta\mu$, then $\{\Lambda_{\mathcal{Q}}(B) \le \theta\mu - \lambda\} \subseteq \{\Lambda_{\mathcal{Q}}(B) < 0\}$ is empty and the bound is trivial; otherwise $\lambda \le \theta\mu$ gives $\frac{2}{3}\mu\lambda \le 2\theta\mu^2$, so the denominator is at most $2\theta\mu(1-\mu) + 2\theta\mu^2 = 2\theta\mu$, yielding Eq (6). □

J. Proof of Lemma 6.2

Write $\mathrm{OPT} = \rho^\diamond(B^o_\diamond)$ and let $\hat{\rho}^\diamond(B) = n^- \Lambda_{\mathcal{Q}}(B)/\theta$ be an unbiased estimator of $\rho^\diamond(B)$. Let $\theta = |\mathcal{Q}_1|$, fix any $0 < \varepsilon_1 < \varepsilon < 1$, and set $\theta_1 = 2n^- \ln(12/\delta)/(\varepsilon_1^2\,\mathrm{OPT})$. When $\theta \ge \theta_1$, the lower tail (6) at the optimum $B^o_\diamond$ gives

$$\Pr\big[\hat{\rho}^\diamond(B^o_\diamond) \le (1-\varepsilon_1)\mathrm{OPT}\big] \;\le\; \exp\Big(-\tfrac{\varepsilon_1^2\,\theta\,\mathrm{OPT}}{2n^-}\Big) \;\le\; \tfrac{\delta}{12}.$$

Given that $\Lambda_{\mathcal{Q}_1}$ is monotone and submodular, the greedy max-coverage gives $\Lambda_{\mathcal{Q}_1}(B_\diamond) \ge (1-1/e)\Lambda_{\mathcal{Q}_1}(B^o_\diamond)$, i.e. $\hat{\rho}^\diamond(B_\diamond) \ge (1-1/e)\hat{\rho}^\diamond(B^o_\diamond)$. Writing $\varepsilon_2 = \varepsilon - (1-1/e)\varepsilon_1$, it follows that, with probability at least $1 - \delta/12$,

$$\begin{aligned}\hat{\rho}^\diamond(B_\diamond) &\ge\; (1-1/e)(1-\varepsilon_1)\,\mathrm{OPT}\\ &=\; (1-1/e-\varepsilon+\varepsilon_2)\,\mathrm{OPT}.\end{aligned}$$

Call a size-k set B *bad* if $\rho^\diamond(B) < (1-1/e-\varepsilon)\,\mathrm{OPT}$, and set $\theta_2 = 2(1-1/e)\,n^- \ln\big(\binom{m}{k}\cdot 12/\delta\big)/(\varepsilon_2^2\,\mathrm{OPT})$. When $\theta \ge \theta_2$, the upper tail (5) applied to any fixed bad B gives

$$\begin{aligned}&\Pr\big[\hat{\rho}^\diamond(B) \ge \rho^\diamond(B) + \varepsilon_2\mathrm{OPT}\big]\\ &\le \exp\Big(-\tfrac{\varepsilon_2^2\,\theta\,\mathrm{OPT}}{\big(2(1-\frac{1}{e}-\varepsilon)+\frac{2}{3}\varepsilon_2\big)n^-}\Big)\\ &\le \tfrac{\delta}{12\binom{m}{k}}.\end{aligned}$$

Union-bounding this over the at most $\binom{m}{k}$ bad sets and intersecting with the high-probability event above, the following two facts hold simultaneously with probability at least $1 - \frac{\delta}{12} - \binom{m}{k}\cdot\frac{\delta}{12\binom{m}{k}} = 1 - \frac{\delta}{6}$:

(a) the greedy solution satisfies: $\hat{\rho}^\diamond(B_\diamond) \ge (1-1/e-\varepsilon+\varepsilon_2)\,\mathrm{OPT}$;

(b) every bad B satisfies: $\hat{\rho}^\diamond(B) < \rho^\diamond(B) + \varepsilon_2\mathrm{OPT} < (1-1/e-\varepsilon+\varepsilon_2)\,\mathrm{OPT}$, where the last step is by definition.

Now suppose, for contradiction, that $B_\diamond$ itself is bad. Then (b) applied with $B = B_\diamond$ gives $\hat{\rho}^\diamond(B_\diamond) < (1-1/e-\varepsilon+\varepsilon_2)\,\mathrm{OPT}$, contradicting (a). Hence $B_\diamond$ is not bad, i.e., $\rho^\diamond(B_\diamond) \ge (1-1/e-\varepsilon)\,\mathrm{OPT}$, with probability at least $1 - \delta/6$.

It remains to collapse the two requirements $\theta \ge \theta_1$ and $\theta \ge \theta_2$ into one. Setting $\theta_1 = \theta_2$ and solving the resulting equation for the split ε_1 makes both equal to the single sufficient size

$$\theta' = \frac{2\,n^-}{\varepsilon^2\,\mathrm{OPT}}\Big((1-1/e)\sqrt{\ln\tfrac{12}{\delta}} + \sqrt{(1-1/e)\big(\ln\tbinom{m}{k} + \ln\tfrac{12}{\delta}\big)}\Big)^2.$$

Since computing OPT is #P-hard, we replace it by the efficiently computable lower bound $\kappa \le \mathrm{OPT}$, which only enlarges the threshold to the $\theta_{\max}$ of Eq. (7). Hence $|\mathcal{Q}_1| \ge \theta_{\max} \ge \theta'$ guarantees $\rho^\diamond(B_\diamond) \ge (1-1/e-\varepsilon)\,\rho^\diamond(B^o_\diamond)$ with probability at least $1 - \delta/6$. □

K. Proof of Lemma 6.3

It suffices to prove that, within each round,

$$\Pr\big[\rho^\diamond(B_\diamond) < \rho^\diamond_\ell(B_\diamond)\big] \le \exp(-\zeta), \tag{13}$$

$$\Pr\big[\rho^\diamond(B^o_\diamond) > \rho^\diamond_u(B^o_\diamond)\big] \le \exp(-\zeta). \tag{14}$$

Since $\exp(-\zeta) = \delta/(12\,i_{\max})$, a union bound over these two events across the at most $i_{\max}$ doubling rounds caps the total failure probability at $2\,i_{\max}\cdot\delta/(12\,i_{\max}) = \delta/6$; hence both bounds hold simultaneously with probability at least $1 - \delta/6$.

Eq (13) is derived from Eq (5). Since $B_\diamond$ is determined by $\mathcal{Q}_1$, it is independent of $\mathcal{Q}_2$, so Eq (5) applies on $\mathcal{Q}_2$. Let $\mu = \rho^\diamond(B_\diamond)/n^-$ and substituting $\rho^\diamond_\ell$ from Eq (8),

$$\begin{aligned}&\Pr\big[\rho^\diamond(B_\diamond) < \rho^\diamond_\ell(B_\diamond)\big]\\ &\le \Pr\Big[\Lambda_{\mathcal{Q}_2}(B_\diamond) - |\mathcal{Q}_2|\mu > \tfrac{\zeta}{3} + \sqrt{2\zeta|\mathcal{Q}_2|\mu + \tfrac{\zeta^2}{9}}\Big] \le \exp(-\zeta),\end{aligned}$$

where the last step applies Eq (5) with $\lambda = \frac{\zeta}{3} + \sqrt{2\zeta|\mathcal{Q}_2|\mu + \zeta^2/9}$, the positive root of $\lambda^2 = \zeta(2|\mathcal{Q}_2|\mu + \frac{2}{3}\lambda)$.

The derivation of Eq (14) is based on Eq (6). Since $B^o_\diamond$ is the deterministic optimum, Eq (6) applies to it on $\mathcal{Q}_1$. We first bound $\Lambda^u_{\mathcal{Q}_1}(B^o_\diamond)$: by the monotonicity and submodularity of $\Lambda_{\mathcal{Q}_1}$, every greedy prefix $B^{(i)}_\diamond$ $(0 \le i \le k)$ satisfies

$$\begin{aligned}\Lambda_{\mathcal{Q}_1}(B^o_\diamond) &\le \Lambda_{\mathcal{Q}_1}\big(B^{(i)}_\diamond \cup B^o_\diamond\big)\\ &\le \Lambda_{\mathcal{Q}_1}(B^{(i)}_\diamond) + \textstyle\sum_{e\in B^o_\diamond}\Lambda_{\mathcal{Q}_1}(e \mid B^{(i)}_\diamond)\\ &\le \Lambda_{\mathcal{Q}_1}(B^{(i)}_\diamond) + \max_{|C|=k}\textstyle\sum_{e\in C}\Lambda_{\mathcal{Q}_1}(e \mid B^{(i)}_\diamond).\end{aligned}$$

Minimizing over i then gives $\Lambda_{\mathcal{Q}_1}(B^o_\diamond) \le \Lambda^u_{\mathcal{Q}_1}(B^o_\diamond)$. Writing $\mu = \rho^\diamond(B^o_\diamond)/n^-$ and substituting $\rho^\diamond_u$ from Eq (9),

$$\begin{aligned}&\Pr\big[\rho^\diamond(B^o_\diamond) > \rho^\diamond_u(B^o_\diamond)\big]\\ &\le \Pr\big[\Lambda^u_{\mathcal{Q}_1}(B^o_\diamond) < |\mathcal{Q}_1|\mu - \sqrt{2\zeta|\mathcal{Q}_1|\mu}\big]\\ &\le \Pr\big[\Lambda_{\mathcal{Q}_1}(B^o_\diamond) < |\mathcal{Q}_1|\mu - \sqrt{2\zeta|\mathcal{Q}_1|\mu}\big]\\ &\le \exp(-\zeta),\end{aligned}$$

where the second inequality uses $\Lambda_{\mathcal{Q}_1}(B^o_\diamond) \le \Lambda^u_{\mathcal{Q}_1}(B^o_\diamond)$ and the last applies Eq (6) with $\lambda = \sqrt{2\zeta|\mathcal{Q}_1|\mu}$. □

L. Proof of Lemma 6.4

The per-node expression of $\rho(B)$ is structurally analogous to those of ρ^L and ρ^U, so the proof follows the same line of reasoning as Lemmas 5.3 and 5.7:

$$\begin{aligned}\rho(B) &= \textstyle\sum_{v\in V\setminus S}\mathbb{E}_\phi\big[\mathbf{1}[v \in I_\phi(S) \wedge v \notin I_{\phi\setminus B}(S)]\big]\\ &= \mathbb{E}_\phi\big[\textstyle\sum_{v\in I_\phi(S)\setminus S}\mathbf{1}[v \notin I_{\phi\setminus B}(S)]\big]\\ &= \mathbb{E}\big[N_\phi\cdot\mathbf{1}[v \notin I_{\phi\setminus B}(S)]\big].\end{aligned}$$

This completes the proof. □

M. Proof of Lemma 6.5

This lemma follows from the theoretical guarantee of the stopping-rule framework [39]: for an i.i.d. stream of $[0,c]$-bounded random variables with mean $\mu > 0$, stopping when the running sum reaches the threshold $2c(1+\gamma)(1+\gamma/3)\ln(2/\delta')/\gamma^2$ and returning this threshold divided

by the corresponding stopping time $\theta^\star$ gives a $(1 \pm \gamma)$-multiplicative estimate of μ with probability at least $1 - \delta'$, using $O(c \ln(1/\delta')/(\gamma^2 \mu))$ samples in expectation. In our case, Lemma 6.4 gives $\mathbb{E}[W(B)] = \rho(B)$, and $W(B) \in [0, n^-]$ by construction. Substituting $c = n^-$, $\mu = \rho(B)$, and $\delta' = \delta/6$ gives the stated accuracy guarantee and expected sample complexity. □

N. Proof of Theorem 6.6

Let $\beta = 1 - 1/e - \varepsilon$. We first show that, with high probability, the two candidates produced by Algorithm 5 are β-approximate for their corresponding surrogate objectives:

$$\rho^\diamond(B_\diamond) \;\geq\; \beta\, \rho^\diamond(B_\diamond^o) \qquad \text{for both } \diamond \in \{L, U\}.$$

For a fixed $\diamond \in \{L, U\}$, Algorithm 5 stops in one of two ways. If the empirical ratio reaches β, then Lemma 6.3 certifies that $\rho^\diamond(B_\diamond)/\rho^\diamond(B_\diamond^o) \geq \alpha^\diamond \geq \beta$, except with probability at most $\delta/6$. Otherwise, the doubling process continues until the sample size reaches the cap $\theta_{\max}$, where Lemma 6.2 ensures $\rho^\diamond(B_\diamond) \geq \beta \rho^\diamond(B_\diamond^o)$, except with probability at most $\delta/6$. A union bound over these two stopping cases shows that, for each fixed $\diamond$, the surrogate approximation event fails with probability at most $\delta/3$. A further union bound over $\diamond \in \{L, U\}$ gives both surrogate guarantees with probability at least $1 - 2\delta/3$.

It remains to account for the final comparison between B_L and B_U. By Lemma 6.5, with probability at least $1 - \delta/3$, the final estimator satisfies, for both $B \in \{B_L, B_U\}$,

$$(1 - \gamma)\rho(B) \leq \hat{\rho}(B) \leq (1 + \gamma)\rho(B).$$

When this estimator event and the two surrogate approximation events all hold, the data-dependent guarantee of the sandwich strategy (Eq. (1)) applies with $\alpha_L = \alpha_U = \beta$, which is exactly the claimed bound. A final union bound over the surrogate event and the estimator event gives success probability at least $1 - \delta$. □

O. Proof of Theorem 6.7

Let $\tau = \rho(B_U)/\rho^U(B_U)$. Condition on the event of Theorem 6.6 together with the one-sided stopping-rule bounds

$$\hat{\rho}(B_U) \leq (1 + \gamma)\rho(B_U) \quad \text{and} \quad \hat{\rho}^U(B_U) \geq (1 - \gamma)\rho^U(B_U).$$

By a union bound, all these events hold simultaneously with probability at least $1 - 2\delta$. Under them, we have

$$\begin{aligned}\tfrac{\rho(B^*)}{\rho(B^o)} &\geq (1 - 1/e - \varepsilon)\, \tfrac{1-\gamma}{1+\gamma}\, \tfrac{\rho(B_U)}{\rho^U(B_U)} \\ &\geq (1 - 1/e - \varepsilon) \left(\tfrac{1-\gamma}{1+\gamma}\right)^2 \tfrac{\hat{\rho}(B_U)}{\hat{\rho}^U(B_U)} = \underline{r}.\end{aligned}$$

which proves the theorem. □

P. Proof of Lemma 6.8

We fix $\diamond \in \{L, U\}$. For simplicity, write $a = 1 - 1/e$, $\mathrm{OPT} = \rho^\diamond(B_\diamond^o)$, $g = \rho^\diamond(B_\diamond)$, and $\hat{\rho}_i^\diamond(B) = \frac{n^-}{\theta} \Lambda_{\mathcal{Q}_i}(B)$ for the estimator on sample collection $\mathcal{Q}_i$ ($i \in \{1, 2\}$). Let $\theta = |\mathcal{Q}_1| = |\mathcal{Q}_2|$ be the running sample size and $\theta^\diamond$ its final value such that $\bar{\theta} = \mathbb{E}[\max\{\theta^L, \theta^U\}]$. To prove the lemma, we argue in three stages. *(a)* We identify four events that jointly imply the stopping condition:

(E1) $\Lambda^u_{\mathcal{Q}_1}(B_\diamond^o) \leq \Lambda_{\mathcal{Q}_1}(B_\diamond)/a$;
(E2) $\hat{\rho}_1^\diamond(B_\diamond) \leq g + \frac{\eta}{3}\mathrm{OPT}$;
(E3) $\hat{\rho}_1^\diamond(B_\diamond^o) \geq (1 - \frac{\eta}{3})\mathrm{OPT}$;
(E4) $\rho_\ell^\diamond(B_\diamond) \geq g - \frac{2\eta}{3}\mathrm{OPT}$,

where the deviation $\eta = \varepsilon/2$ satisfies $\eta < a/2 < 0.32$. *(b)* We fix a unit sample size $\theta' = \max\{\theta_a, \theta_b\}$, where

$$\theta_a = \frac{(18 + 2\eta)\, n^-}{\eta^2\, \mathrm{OPT}} \ln \frac{8\binom{m}{k}}{\delta},$$
$$\theta_b = \frac{72\, n^-\, \zeta}{\eta^2\, \mathrm{OPT}}, \quad \zeta = \ln \frac{12\, i_{\max}}{\delta}$$

and show that once $\theta \geq \alpha\theta'$ for any $\alpha \geq 1$, all four events hold simultaneously with probability at least $1 - \delta^\alpha$. *(c)* We convert this decaying failure into the expected sample size.

It can be seen that θ_a and θ_b are both $O\big(n^- \ln(\binom{m}{k})/\delta)/(\varepsilon^2 \mathrm{OPT})\big)$: clearly for θ_a, and for θ_b we bound $\zeta = \ln(12 i_{\max}/\delta)$. Recall $i_{\max} = \lceil \log_2(n^-/(\varepsilon^2\kappa)) \rceil + 1$. Under the standard assumption of $\mathrm{poly}(n)$-bit edge probabilities, the factor $\kappa = \sum_{v \in T_k} q_v$ satisfies $\kappa \geq 2^{-\mathrm{poly}(n)}$, since each $q_v = p_{u,v} \prod_w (1 - p_{w,v})$ is a product of at most n probabilities, each of which is at least $2^{-\mathrm{poly}(n)}$. Consequently $\log_2 \frac{1}{\kappa} = \mathrm{poly}(n)$, and

$$\ln i_{\max} = O\big(\log\log(n^-/(\varepsilon^2\kappa))\big) = O\big(\log n + \log\log \tfrac{1}{\varepsilon}\big).$$

Both terms are dominated by $\ln \binom{m}{k}$. The first is immediate. The second follows from $\varepsilon \geq 2^{-\binom{m}{k}}$, which gives $\log\log \frac{1}{\varepsilon} = O(\ln \binom{m}{k})$. Hence $\zeta = O(\ln \binom{m}{k} + \ln \frac{1}{\delta})$ and $\theta_b = O\big(n^- \ln(\binom{m}{k})/\delta)/(\varepsilon^2 \mathrm{OPT})\big)$.

Derivation of (E1). Let $B_\diamond^{(0)}, \ldots, B_\diamond^{(k)} = B_\diamond$ be the greedy prefixes and $d_i = \max_{|C|=k} \sum_{e \in C} \Lambda_{\mathcal{Q}_1}(e \mid B_\diamond^{(i)})$ the best-k marginal gain at $B_\diamond^{(i)}$. By definition, $\Lambda^u_{\mathcal{Q}_1}(B_\diamond^o) = \min_{0 \leq i \leq k}(\Lambda_{\mathcal{Q}_1}(B_\diamond^{(i)}) + d_i) \leq \Lambda_{\mathcal{Q}_1}(B_\diamond^{(i)}) + d_i$ for every i. Greedy adds the single element of largest marginal, whose gain is at least the average d_i/k, so $\Lambda_{\mathcal{Q}_1}(B_\diamond^{(i+1)}) - \Lambda_{\mathcal{Q}_1}(B_\diamond^{(i)}) \geq d_i/k \geq (\Lambda^u_{\mathcal{Q}_1}(B_\diamond^o) - \Lambda_{\mathcal{Q}_1}(B_\diamond^{(i)}))/k$; equivalently,

$$\begin{aligned}&\Lambda^u_{\mathcal{Q}_1}(B_\diamond^o) - \Lambda_{\mathcal{Q}_1}(B_\diamond^{(i+1)}) \\ &\leq (1 - \tfrac{1}{k})\big(\Lambda^u_{\mathcal{Q}_1}(B_\diamond^o) - \Lambda_{\mathcal{Q}_1}(B_\diamond^{(i)})\big).\end{aligned}$$

Iterating from $\Lambda_{\mathcal{Q}_1}(B_\diamond^{(0)}) = 0$,

$$\Lambda^u_{\mathcal{Q}_1}(B_\diamond^o) - \Lambda_{\mathcal{Q}_1}(B_\diamond) \leq (1 - \tfrac{1}{k})^k\, \Lambda^u_{\mathcal{Q}_1}(B_\diamond^o) \leq \tfrac{1}{e}\, \Lambda^u_{\mathcal{Q}_1}(B_\diamond^o),$$

so $\Lambda_{\mathcal{Q}_1}(B_\diamond) \geq (1 - \frac{1}{e})\Lambda^u_{\mathcal{Q}_1}(B_\diamond^o)$, which is (E1).

Derivation of (E2). This is an upper-tail event on the selection pool $\mathcal{Q}_1$, required over all size-k sets. It permits the absolute deviation $\frac{\eta}{3}\mathrm{OPT}$ for every B, and we apply Eq. (5) with

$$\mu = \rho^\diamond(B)/n^-, \qquad \lambda = \theta\, \tfrac{\eta}{3}\mathrm{OPT}/n^-.$$

It follows that, for each fixed size-k B,

$$\Pr\big[\hat{\rho}_1^\diamond(B) > \rho^\diamond(B) + \tfrac{\eta}{3}\mathrm{OPT}\big] \;\leq\; \exp\Big(-\tfrac{\lambda^2}{2\theta\mu + \frac{2}{3}\lambda}\Big).$$

Since λ does not depend on B, the right-hand side increases with μ, and is thus largest at the maximum feasible $\rho^\diamond(B) = \mathrm{OPT}$, i.e. at $\mu = \mathrm{OPT}/n^-$. It holds that

$$\frac{\lambda^2}{2\theta\mu+\frac{2}{3}\lambda} = \frac{\eta^2\,\theta\,\mathrm{OPT}}{(18+2\eta)\,n^-} \;\geq\; \alpha \ln\frac{8\binom{m}{k}}{\delta} \qquad \text{whenever } \theta \geq \alpha\theta_a,$$

Each fixed B therefore fails with probability at most $(8\binom{m}{k}/\delta)^{-\alpha}$, and a union bound over the $\binom{m}{k}$ size-k sets gives

$$\Pr[\neg\mathrm{E2}] \;\leq\; \tbinom{m}{k}\Big(\frac{8\binom{m}{k}}{\delta}\Big)^{-\alpha} \;\leq\; \Big(\frac{\delta}{8}\Big)^{\alpha}.$$

Derivation of (E3). This is a lower-tail event at the single fixed set $B^o_\diamond$, thus no union bound is required. To start, by substituting $\lambda = \beta\theta\mu$ into Eq. (6), we have

$$\Pr[\hat\rho^\diamond_i(B) \leq (1-\beta)\rho^\diamond(B)] \;\leq\; e^{-\beta^2\theta\rho^\diamond(B)/(2n^-)}. \tag{15}$$

Applying it at $B^o_\diamond$ with $\beta = \frac{\eta}{3}$, and using $e^{-\zeta} = \delta/(12 i_{\max})$,

$$\begin{aligned}\Pr[\neg\mathrm{E3}] &\leq \exp\Big(-\tfrac{\eta^2\,\theta\,\mathrm{OPT}}{18\,n^-}\Big)\\ &\leq \Big(\tfrac{\delta}{12\,i_{\max}}\Big)^{\alpha} \text{ whenever } \theta \geq \alpha\theta_b.\end{aligned}$$

Derivation of (E4). This is the validation event on the fresh pool $\mathcal{Q}_2$. Assuming that (E2) and (E3) holds. Rearranging (E2), then applying the greedy guarantee $\Lambda_{\mathcal{Q}_1}(B_\diamond) \geq a\,\Lambda_{\mathcal{Q}_1}(B^o_\diamond)$, and then (E3),

$$\begin{aligned}g &\geq\; \hat\rho^\diamond_1(B_\diamond) - \tfrac{\eta}{3}\mathrm{OPT} \;\geq\; a\,\hat\rho^\diamond_1(B^o_\diamond) - \tfrac{\eta}{3}\mathrm{OPT}\\ &\geq\; a(1-\tfrac{\eta}{3})\mathrm{OPT} - \tfrac{\eta}{3}\mathrm{OPT} \;=\; \Big(a - \tfrac{a+1}{3}\eta\Big)\mathrm{OPT}.\end{aligned} \tag{16}$$

Since $\frac{a+1}{3} < 1$, it implies $g \geq (a-\eta)\mathrm{OPT}$; with $\eta < 0.32$ and $a > 0.63$ this gives $g \geq 0.31\,\mathrm{OPT}$.

(E4) concerns $\rho^\diamond_\ell(B_\diamond)$, which by Eq. (8) is a deterministic function of the coverage $\Lambda_{\mathcal{Q}_2}(B_\diamond)$ that $\mathcal{Q}_2$ reports:

$$\rho^\diamond_\ell(B_\diamond) = \tfrac{n^-}{\theta}\,f\big(\Lambda_{\mathcal{Q}_2}(B_\diamond)\big),$$
$$f(x) = \Big(\sqrt{x+\tfrac{2\zeta}{9}} - \sqrt{\tfrac{\zeta}{2}}\Big)^2 - \tfrac{\zeta}{18}.$$

We therefore prove (E4) in two steps: bound the coverage from below with high probability, then push that bound through f.

Step 1: a lower bound on the coverage. Apply Eq. (15) at $B_\diamond$ on $\mathcal{Q}_2$ with $\beta = \frac{\eta}{3}$. Writing $L = (1-\frac{\eta}{3})\theta g/n^-$, we get $\Lambda_{\mathcal{Q}_2}(B_\diamond) \geq L$ except with probability

$$\begin{aligned}\exp\Big(-\tfrac{\eta^2\theta g}{18n^-}\Big) &\leq\; e^{-4\alpha\zeta\,g/\mathrm{OPT}}\\ &\leq\; \Big(\tfrac{\delta}{12\,i_{\max}}\Big)^{\alpha} \quad \text{whenever } \theta \geq \alpha\theta_b,\end{aligned}$$

the last inequality using $g \geq 0.31\,\mathrm{OPT} > \frac{1}{4}\mathrm{OPT}$. Then, by the facts $\theta \geq \theta_b$, $g \geq 0.31\,\mathrm{OPT}$ and $\eta < 0.32$, we have

$$L \;\geq\; (1-\tfrac{\eta}{3})\cdot\tfrac{72\,\zeta}{\eta^2}\cdot\tfrac{g}{\mathrm{OPT}} \;>\; \big(1-\tfrac{0.32}{3}\big)\cdot\tfrac{72}{0.32^2}\cdot 0.31\,\zeta \geq \tfrac{\zeta}{2}.$$

Step 2: pushing the bound through f. The derivative of f is $f'(x) = 1 - \sqrt{\frac{\zeta}{2}}/\sqrt{x+\frac{2\zeta}{9}}$, which is non-negative exactly when $x \geq \frac{5\zeta}{18}$. Since $\Lambda_{\mathcal{Q}_2}(B_\diamond) \geq L \geq \frac{\zeta}{2} > \frac{5\zeta}{18}$, the function f is increasing on $[L,\infty)$, so Step 1 gives $\rho^\diamond_\ell(B_\diamond) \geq \frac{n^-}{\theta}f(L)$. To simplify $f(L)$ we replace $\sqrt{L+\frac{2\zeta}{9}}$ by the smaller $\sqrt{L}$; this can only decrease the square, because $L \geq \frac{\zeta}{2}$ makes both $\sqrt{L+\frac{2\zeta}{9}} - \sqrt{\frac{\zeta}{2}}$ and $\sqrt{L} - \sqrt{\frac{\zeta}{2}}$ non-negative. Hence

$$\begin{aligned}\rho^\diamond_\ell(B_\diamond) &\geq\; \tfrac{n^-}{\theta}f(L) \;\geq\; \tfrac{n^-}{\theta}\Big[\big(\sqrt{L}-\sqrt{\tfrac{\zeta}{2}}\big)^2 - \tfrac{\zeta}{18}\Big]\\ &\geq\; \tfrac{n^-}{\theta}\big(L - \sqrt{2L\zeta}\big)\\ &=\; (1-\tfrac{\eta}{3})\,g - \tfrac{n^-}{\theta}\sqrt{2L\zeta}.\end{aligned}$$

It remains to bound the residual term $\frac{n^-}{\theta}\sqrt{2L\zeta}$. Substituting $L = (1-\frac{\eta}{3})\theta g/n^-$, and then $1-\frac{\eta}{3} \leq 1$ and $g \leq \mathrm{OPT}$,

$$\tfrac{n^-}{\theta}\sqrt{2L\zeta} \;=\; \sqrt{\tfrac{2\zeta(1-\frac{\eta}{3})\,g\,n^-}{\theta}} \;\leq\; \sqrt{\tfrac{2\zeta\,\mathrm{OPT}\,n^-}{\theta}} \;\leq\; \tfrac{\eta}{3}\mathrm{OPT},$$

where the last inequality holds precisely when $\theta \geq 18\zeta n^-/(\eta^2\mathrm{OPT})$, which $\theta \geq \theta_b = 72\zeta n^-/(\eta^2\mathrm{OPT})$ guarantees. Substituting this back and using $g \leq \mathrm{OPT}$, we have

$$\begin{aligned}\rho^\diamond_\ell(B_\diamond) &\geq\; (1-\tfrac{\eta}{3})g - \tfrac{\eta}{3}\mathrm{OPT} \;=\; g - \tfrac{\eta}{3}g - \tfrac{\eta}{3}\mathrm{OPT}\\ &\geq\; g - \tfrac{2\eta}{3}\mathrm{OPT},\end{aligned}$$

which is (E4). As Step 2 is deterministic, (E4) can fail only if Step 1 does; hence, at $\theta \geq \alpha\theta_b$,

$$\Pr[\neg\mathrm{E4}\wedge\mathrm{E2}\wedge\mathrm{E3}] \;\leq\; \big(\tfrac{\delta}{12\,i_{\max}}\big)^{\alpha}. \tag{17}$$

From the four events to the stopping condition. Fix $\theta \geq \alpha\theta'$. Combining (E1)–(E4) by the union bound,

$$\begin{aligned}&\Pr[\neg(\mathrm{E1}\wedge\mathrm{E2}\wedge\mathrm{E3}\wedge\mathrm{E4})]\\ &\leq\; \Pr[\neg\mathrm{E2}] + \Pr[\neg\mathrm{E3}] + \Pr[\neg\mathrm{E4}\wedge\mathrm{E2}\wedge\mathrm{E3}]\\ &\leq\; (\delta/8)^\alpha + 2(\delta/12i_{\max})^\alpha \;\leq\; \delta^\alpha.\end{aligned}$$

Assume these four events hold, and abbreviate $A = \Lambda^u_{\mathcal{Q}_1}(B^o_\diamond)$.

We first bound $\rho^\diamond_u(B^o_\diamond)$, showing that it exceeds $\frac{n^-}{\theta}A$ by a factor of at most $1+0.4\eta$. By the identity $\Lambda_{\mathcal{Q}_1}(B^o_\diamond) \leq A$, then (E3), and then $\theta \geq \theta_b$ together with $1-\frac{\eta}{3} \geq \frac{2}{3}$,

$$A \;\geq\; \Lambda_{\mathcal{Q}_1}(B^o_\diamond) \;\geq\; (1-\tfrac{\eta}{3})\tfrac{\theta\,\mathrm{OPT}}{n^-} \;\geq\; \tfrac{2}{3}\cdot\tfrac{72\,\zeta}{\eta^2} \;=\; \tfrac{48\,\zeta}{\eta^2},$$

so the ratio $t = \zeta/A$ satisfies $t \leq \eta^2/48 \leq 1$. Since $A + \frac{\zeta}{2} = A(1+\frac{t}{2})$ and $\frac{\zeta}{2} = A\frac{t}{2}$, both square roots in Eq. (9) carry a factor $\sqrt{A}$, and pulling it out gives

$$\begin{aligned}\rho^\diamond_u(B^o_\diamond) &= \tfrac{n^-}{\theta}\Big(\sqrt{A+\tfrac{\zeta}{2}} + \sqrt{\tfrac{\zeta}{2}}\Big)^2 = \tfrac{n^-}{\theta}A\Big(\sqrt{1+\tfrac{t}{2}} + \sqrt{\tfrac{t}{2}}\Big)^2\\ &= \tfrac{n^-}{\theta}A\Big(\big(1+\tfrac{t}{2}\big) + \tfrac{t}{2} + 2\sqrt{\big(1+\tfrac{t}{2}\big)\tfrac{t}{2}}\Big),\end{aligned}$$

Since $t \leq 1$, we have $t \leq \sqrt{t}$ and $\sqrt{1+\frac{t}{2}} \leq \sqrt{1.5}$, so

$$\begin{aligned}t + 2\sqrt{\big(1+\tfrac{t}{2}\big)\tfrac{t}{2}} &\leq\; \sqrt{t} + 2\sqrt{1.5}\cdot\sqrt{\tfrac{t}{2}} \;=\; \big(1+\sqrt{3}\big)\sqrt{t}\\ &\leq\; 2.74\sqrt{t} \;\leq\; \tfrac{2.74}{\sqrt{48}}\,\eta \;<\; 0.4\,\eta,\end{aligned}$$

the last two steps using $t \leq \eta^2/48$. Altogether

$$\rho^\diamond_u(B^o_\diamond) \;\leq\; (1+0.4\,\eta)\,\tfrac{n^-}{\theta}A. \tag{18}$$

Combining (E1) and (E2) gives $\frac{n^-}{\theta}A \le \frac{1}{a}\hat{\rho}_1^\diamond(B_\diamond) \le \frac{1}{a}(g+\frac{\eta}{3}\mathrm{OPT})$, so Eq. (18) yields $\rho_u^\diamond(B_\diamond^o) \le (1+0.4\eta)(g+\frac{\eta}{3}\mathrm{OPT})/a$. Dividing (E4) by this, we have

$$\alpha_\diamond = \frac{\rho_\ell^\diamond(B_\diamond)}{\rho_u^\diamond(B_\diamond^o)} \;\ge\; \frac{a}{1+0.4\,\eta}\cdot\frac{g-\frac{2\eta}{3}\mathrm{OPT}}{g+\frac{\eta}{3}\mathrm{OPT}}.$$

The last fraction is increasing in g, hence smallest at the lower bound of Eq. (16), namely $g=(a-\frac{a+1}{3}\eta)\mathrm{OPT}$:

$$a\cdot\frac{a-\frac{a\eta}{3}-\eta}{a-\frac{a\eta}{3}} = a\cdot\frac{a(1-\frac{\eta}{3})-\eta}{a(1-\frac{\eta}{3})} = a-\frac{\eta}{1-\frac{\eta}{3}}.$$

Therefore, using $\frac{1}{1+0.4\eta}\ge 1-0.4\eta$, then $a<1$, and finally $\eta<0.32$ (so $\frac{1}{1-\eta/3}\le 1.12$), it holds that

$$\begin{aligned}\alpha_\diamond &\ge\; (1-0.4\,\eta)\Big(a-\tfrac{\eta}{1-\eta/3}\Big) \;\ge\; a-\tfrac{\eta}{1-\eta/3}-0.4\,\eta\,a\\ &\ge\; a-1.12\,\eta-0.4\,\eta = a-1.52\,\eta\\ &=\; a-0.76\,\varepsilon \;\ge\; a-\varepsilon.\end{aligned}$$

Thus, once $\theta\ge\alpha\theta'$, the stopping condition $\alpha_\diamond\ge 1-1/e-\varepsilon$ holds, except with probability δ^α.

Expected sample size. SABRE doubles each pool, so its size at iteration $z=0,1,\dots$ of the loop is $\theta_0 2^z$. A pool stops growing once its stopping condition holds or the last of the $i_{\max}$ iterations is reached. Let j be the first iteration at which the pool size reaches θ'. From this iteration onward, the expected number of samples further drawn is at most

$$\begin{aligned}2\cdot\sum_{z\ge j}\theta_0 2^z\,\delta^{2^{z-j}} \;&=\; 2\cdot 2^j\theta_0\sum_{z\ge 0}2^z\,\delta^{2^z}\\ \le\; 4\theta'\sum_{z\ge 0}2^{-2^z+z} \;&\le\; 4\theta'\sum_{z\ge 0}2^{-z} \;\le\; 8\theta'.\end{aligned}$$

The first inequality is due to $2^{j-1}\theta_0<\theta'$ and $\delta\le\frac{1}{2}$, and the second inequality is due to $-2^z+z\le -z$. If the pool stops growing earlier, at most $2\theta'$ samples are drawn. Hence, the expected number of samples drawn is less than $10\theta'$, i.e.,

$$\mathbb{E}[\theta^\diamond] < 10\,\theta' \;=\; O\big(n^-\ln(\tbinom{m}{k}/\delta)/(\varepsilon^2\mathrm{OPT})\big).$$

Finally, a single ForwardSample feeds both bounds and each pool ceases doubling once its stopping condition holds, so the number of joint calls equals $\max\{\theta^L,\theta^U\}$. Since $\rho^U(B)\ge\rho^L(B)$ holds for every B, we have $\rho^U(B_U^o)\ge\rho^U(B_L^o)\ge\rho^L(B_L^o)$. As a result,

$$\begin{aligned}\bar\theta=\mathbb{E}[\max\{\theta^L,\theta^U\}] &\le\; \mathbb{E}[\theta^L]+\mathbb{E}[\theta^U]\\ &=\; O\big(n^-\ln(\tbinom{m}{k}/\delta)/(\varepsilon^2\rho^L(B_L^o))\big).\end{aligned}$$

The proof is completed. □

Q. Proof of Theorem 6.9

We decompose the running time into the costs of sample generation, greedy selection, stopping-condition evaluation, and final objective estimation, denoted by C_{gen}, C_{sel}, C_{test}, and C_{est}, respectively. At doubling iteration z, each pool contains $\theta_z=\theta_0 2^z$ samples, where $\theta_0=\theta_{\max}\varepsilon^2\kappa/n^-$, and the loop has at most $i_{\max}$ iterations.

Sample generation. A joint sample is generated by a single ForwardSample call, whose forward BFS over G costs $O(m)$ and serves both the CE- and AE-set constructions. The CE set is extracted from the targeted reachability subgraph $H_{\phi,v}$ (Definition 5.4), with vertex set V_H and edge set E_H, in three steps: a reverse BFS from the target v builds $H_{\phi,v}$ in $O(|E_H|)$ time, its dominator tree is constructed in $O(|E_H|\log|V_H|)$ time, and a single scan recovers the CE set in $O(|E_H|)$ time, for a total of $O(|E_H|\log|V_H|)$. The AE set is cheaper, as it only traces the unique S-to-v path in the reachability forest, costing $O(|E_H|)$. Summing these costs and using $|E_H|\le m$ and $|V_H|\le n$, each joint sample costs $O(m\log n)$. Since the doubling loop draws $\bar\theta$ joint samples in expectation (Lemma 6.8), Wald's equation gives $\mathbb{E}[C_{\mathrm{gen}}]=O(m\log n\cdot\bar\theta)$.

Greedy selection. At iteration z, each sample collection has total size $O(m\theta_z)$, over which weighted greedy makes k marginal-gain passes at a cost of $O(km\theta_z)$. As the pool sizes grow geometrically, the cumulative cost for each bound is dominated by its final pool, of size θ^L for the lower bound and θ^U for the upper. Summing the two and using $\theta^L+\theta^U\le 2\max\{\theta^L,\theta^U\}$, we have

$$\mathbb{E}[C_{\mathrm{sel}}] = O\big(km\,\mathbb{E}[\theta^L+\theta^U]\big) = O(km\bar\theta).$$

Stopping test. At iteration z, the test evaluates a lower and an upper certificate. The lower certificate $\rho_\ell^\diamond(B_\diamond)$ is obtained from a single coverage pass over the validation collection, costing $O(m\theta_z)$. The upper certificate $\rho_u^\diamond(B_\diamond^o)$ is obtained from

$$\Lambda_{\mathcal{Q}_1}^u(B_\diamond^o)=\min_{0\le i\le k}\Big(\Lambda_{\mathcal{Q}_1}(B_\diamond^{(i)})+\max_{|C|=k}\sum_{e\in C}\Lambda_{\mathcal{Q}_1}(e\mid B_\diamond^{(i)})\Big),$$

which reuses the marginal gains from greedy and, at each of the $O(k)$ prefixes, sums their top k values in $O(km)$ time overall. The test therefore costs $O(m\theta_z+km)=O(km\theta_z)$ per iteration, and summing over the geometric schedule gives $\mathbb{E}[C_{\mathrm{test}}]=O(km\bar\theta)$.

Final objective estimation. Condition on the candidates B_L and B_U returned by the loop, and let T_B be the stopping time of ObjectiveEstimation for B. Lemma 6.5 gives

$$\mathbb{E}[T_B\mid B]=O\left(\frac{n^-\ln(1/\delta)}{\gamma^2\rho(B)}\right).$$

The procedure stops once both candidates cross the threshold, after $\max\{T_{B_L},T_{B_U}\}$ draws, each evaluating the two candidates by reverse reachability searches in $O(m)$ time. Hence,

$$\mathbb{E}[C_{\mathrm{est}}\mid B_L,B_U]=O\left(\frac{n^-m\ln(1/\delta)}{\gamma^2\min\{\rho(B_L),\rho(B_U)\}}\right).$$

Summing the loop costs C_{gen}, C_{sel}, C_{test} (with $\bar\theta$ bounded by Lemma 6.8) and the estimation cost C_{est} yields the two terms of Theorem 6.9, completing the proof. □